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Aircraft comprising at least one engine attachment system that has reduced vertical bulk

Abstract

An aircraft comprising at least one engine attachment system connecting a primary structure of a pylon and a motor, having a right-hand interface connected to the right-hand lateral face of the primary structure, a left-hand interface connected to the left-hand lateral face of the primary structure, first and second shackles connecting the right-hand and left-hand interfaces to the motor by virtue of engine connection pins and interface connection pins, a right-hand emergency connection pin cooperating with the right-hand interface and the motor, and a left-hand emergency connection pin cooperating with the left-hand interface and the motor.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) This application claims the benefit of French Patent Application Number 2310696 filed on Oct. 6, 2023, the entire disclosures of which are incorporated herein by way of reference.

FIELD OF THE INVENTION

(2) The present application relates to an aircraft comprising at least one engine attachment system that has reduced vertical bulk.

BACKGROUND OF THE INVENTION

(3) According to an embodiment visible in FIGS. **1** and **2**, an aircraft **10** comprises a plurality of propulsion assemblies **12** positioned beneath each of the wings **14** of the aircraft. Each propulsion assembly **12** comprises a motor **16**, a nacelle (not shown in FIG. **2**) positioned around the motor **16**, and a pylon **18** connecting the motor **16** and the wing **14**. The pylon **18** comprises a primary structure **20** that is connected to the motor **16** by an engine attachment system **22** and to the wing **14** by a wing attachment system **24**.

(4) For the present invention, a longitudinal direction X is substantially parallel to the axis of rotation **A16** of the motor **16**. A vertical median plane PMV is a vertical plane passing through the axis of rotation **A16**. A transverse plane is a plane perpendicular to the axis of rotation **A16**. A horizontal transverse direction Y is a horizontal direction perpendicular to the axis of rotation **A16**. A vertical direction is denoted Z in the various figures. The terms front/rear refer to the direction of

flow of the airflows that flow in the longitudinal direction X, from the front toward the rear.

(5) The engine attachment system **22** comprises a front engine attachment system **26**, a rear engine attachment system **28** and a pair of thrust rods **30** that react the thrust forces.

(6) According to an embodiment visible in FIG. 3, for the rear engine attachment system **28**, the primary structure **20** comprises a sole **32** positioned approximately in a horizontal plane.

(7) In addition, the rear engine attachment system **28** comprises a transverse beam **34** that has a bearing face **F34** pressed against the sole **32**, the transverse beam **34** being connected to the primary structure **20** by connection elements **36**.

(8) The rear engine attachment system **28** comprises: a first, L-shaped shackle **38**, which has a first end **38.1**, a second end **38.2** and an intermediate zone **38.3**, a first engine connection pin **40** connecting the first end **38.1** of the first shackle **38** and the motor **16**, a first beam connection pin **42** connecting the second end **38.2** of the first shackle **38** and the transverse beam **34**, a second beam connection pin **44** connecting the intermediate zone **38.3** of the first shackle **38** and the transverse beam **34**; a second shackle **46**, which has first and second ends **46.1**, **46.2**, a second engine connection pin **48** connecting the first end **46.1** of the second shackle **46** and the motor **16**, a third beam connection pin **50** connecting the second end **46.2** of the second shackle **46** and the transverse beam **34**; an emergency connection pin **52**, directly connecting the transverse beam **34** and the motor **16** in the event of a failure, positioned at the vertical median plane PMV.

(9) The connection pins **40**, **42**, **44**, **48**, **50**, **52** are substantially parallel to each other and to the longitudinal direction X.

(10) According to the embodiment visible in FIG. 3, the transverse beam **34** is necessarily positioned beneath the primary structure **20** of the pylon **18**. This arrangement leads to significant bulk in the vertical direction, and this tends to limit the diameter of the motor **16** and more particularly of its fan casing **16.1**.

(11) The present invention aims to remedy all or some of the drawbacks of the prior art.

SUMMARY OF THE INVENTION

(12) To this end, the invention relates to an aircraft comprising at least one motor, at least one primary structure of a pylon and at least one engine attachment system connecting the primary structure and the motor, the primary structure having a lower face and right-hand and left-hand lateral faces positioned on either side of a vertical median plane, the engine attachment system having: at least one first, L-shaped shackle, positioned in a transverse plane, which has first and second ends and an intermediate zone, a first engine connection pin connecting the first end of the first shackle and the motor, at least one second shackle, positioned in a transverse plane, which has first and second ends, a second engine connection pin connecting the first end of the second shackle and the motor.

(13) According to the invention, the engine attachment system comprises: a right-hand interface connected to the right-hand lateral face of the primary structure, a left-hand interface connected to the left-hand lateral face of the primary structure, a first interface connection pin connecting the second end of the first shackle and a first interface among the right-hand and left-hand interfaces, a second interface connection pin connecting the intermediate zone of the first shackle and the first interface, a third interface connection pin connecting the second end of the second shackle and a second interface, which is different from the first interface, among the right-hand and left-hand interfaces, a right-hand emergency connection pin cooperating with the right-hand interface and the motor, a left-hand emergency connection pin cooperating with the left-hand interface and the motor.

(14) According to this arrangement, since the right-hand and left-hand interfaces are not positioned beneath the primary structure but on the lateral faces thereof, the engine attachment system is relatively compact in the vertical direction, and this makes it possible to install motors of larger diameters.

(15) According to another feature, the first, second and third interface connection pins are situated

at a lower plane passing through the lower face of the primary structure or offset upward with respect to this lower plane.

(16) According to another feature, the first interface among the right-hand and left-hand interfaces comprises, for each of the first and second interface connection pins, at least one through-orifice housing the first or second interface connection pin and dimensioned such that the first or second interface connection pin is immobile in translation, in a transverse plane, with respect to the first interface. In parallel, the second interface among the right-hand and left-hand interfaces comprises at least one through-orifice housing the third interface connection pin and dimensioned such that the third interface connection pin is immobile in translation, in a transverse plane, with respect to the second interface.

(17) According to another feature, each right-hand or left-hand interface comprises front and rear arms positioned in transverse planes, parallel to each other and slightly spaced apart, and a right-hand or left-hand base connecting the front and rear arms, the right-hand or left-hand base being connected to the primary structure by at least one connection element.

(18) According to another feature, the engine attachment system comprises at least one reinforcing bracket having a first wall pressed against one of the front and rear arms of the right-hand and left-hand interfaces and a second wall pressed against the right-hand or left-hand lateral face of the primary structure and connected to the primary structure by at least one connection element; the first wall having, for each of the first, second and third interface connection pins passing through it, a through-orifice dimensioned such that the first, second or third interface connection pin is immobile in translation, in a transverse plane, with respect to the first wall.

(19) According to another feature, for each right-hand and left-hand interface, the engine attachment system comprises a front reinforcing bracket, positioned at the front of the right-hand or left-hand interface, having a first wall pressed against the front arm of the right-hand or left-hand interface and a second wall pressed against the right-hand or left-hand lateral face of the primary structure and a rear reinforcing bracket, positioned at the rear of the right-hand or left-hand interface, having a first wall pressed against the rear arm of the right-hand or left-hand interface and a second wall pressed against the right-hand or left-hand lateral face of the primary structure.

(20) According to another feature, each front and rear arm of each right-hand and left-hand interface comprises a side oriented toward the motor and an extension positioned at the side and offset toward the motor with respect to the remainder of the side, the extension having a through-orifice that houses the right-hand or left-hand emergency connection pin.

(21) According to another feature, the motor comprises, for each right-hand or left-hand emergency connection pin, a transverse flange positioned at least partially between the front and rear arms of the corresponding right-hand or left-hand interface, the transverse flange having a through-orifice that houses the right-hand or left-hand emergency connection pin.

(22) According to another feature, the motor comprises, for the first engine connection pin, a first clevis that has two arms between which the first shackle is positioned and, for the second engine connection pin, a second clevis that has two arms between which the second shackle is positioned.

(23) According to another feature, the engine attachment system comprises, for each right-hand or left-hand emergency connection pin, a clearance between the right-hand or left-hand emergency connection pin and one element among the motor and the right-hand or left-hand interface.

(24) According to another feature, the motor comprises, for each of the first and second engine connection pins, at least one through-orifice, which houses the first or second engine connection pin, dimensioned such that each of the first and second engine connection pins is immobile in translation, in a transverse plane, with respect to the motor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Further features and advantages will become apparent from the following description of the invention, which description is given solely by way of example, with reference to the appended drawings in which:
- (2) FIG. 1 is a side view of an aircraft,
- (3) FIG. 2 is a side view of a motor,
- (4) FIG. 3 is a front view of an engine attachment system, which illustrates an embodiment of the prior art,
- (5) FIG. 4 is a perspective view of an engine attachment system, which illustrates an embodiment of the invention,
- (6) FIG. 5 is a front view of the engine attachment system visible in FIG. 4,
- (7) FIG. 6 is a side view of the engine attachment system visible in FIG. 4,
- (8) FIG. 7 is a front view of the engine attachment system visible in FIG. 4, illustrating a failure at one of the shackles,
- (9) FIGS. 8A, 8B and 8C depict side views of the engine attachment system visible in FIG. 4, illustrating various failures.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (10) An aircraft comprises at least one motor **56**, at least one primary structure **58** of a pylon, and at least one engine attachment system **60** connecting the motor **56** and the primary structure **58**. According to one arrangement, this engine attachment system **60**, referred to as rear engine attachment system, is moved away from the fan casing of the motor **56**.
- (11) The motor **56** has a vertical median plane PMV and a substantially cylindrical casing in line with the engine attachment system **60**.
- (12) According to an embodiment visible in FIGS. 4 to 8C, the primary structure **58** comprises a lower face **62** substantially perpendicular to the vertical median plane PMV, and right-hand and left-hand lateral faces **64.1**, **64.2** positioned on either side of the vertical median plane PMV and above the lower face **62**. According to one configuration, each right-hand or left-hand lateral face **64.1**, **64.2** forms, in a transverse plane, an angle of between 0 and 20° with the vertical median plane PMV, the right-hand and left-hand lateral faces **64.1**, **64.2** being flared upward.
- (13) The engine attachment system **60** comprises: a right-hand interface **66** connected to the right-hand lateral face **64.1** of the primary structure **58**, a left-hand interface **68** connected to the left-hand lateral face **64.2** of the primary structure **58**, at least one first, L-shaped shackle **70**, positioned in a transverse plane, which has a first end **70.1**, a second end **70.2** and an intermediate zone **70.3** situated at the bent zone of the first, L-shaped shackle **70**, a first engine connection pin **72** connecting the first end **70.1** of the first shackle **70** and the motor **56**, a first interface connection pin **74** connecting the second end **70.2** of the first shackle **70** and a first interface among the right-hand and left-hand interfaces **66**, **68**, a second interface connection pin **76** connecting the intermediate zone **70.3** of the first shackle **70** and the first interface; at least one second shackle **78**, positioned in a transverse plane, which has first and second ends **78.1**, **78.2**, a second engine connection pin **80** connecting the first end **78.1** of the second shackle **78** and the motor **56**, a third interface connection pin **82** connecting the second end **78.2** of the second shackle **78** and a second interface among the right-hand and left-hand interfaces **66**, **68**; a right-hand emergency connection pin **84** cooperating with the right-hand interface **66** and the motor **56**, a left-hand emergency connection pin **86** cooperating with the left-hand interface **68** and the motor **56**.
- (14) The engine connection, interface connection and emergency connection pins are parallel to each other and to the longitudinal direction X.
- (15) According to this arrangement, the right-hand and left-hand interfaces **66**, **68** are two distinct and dissociated elements. Since these right-hand and left-hand interfaces **66**, **68** are not positioned beneath the primary structure **58** but on the lateral faces thereof, the engine attachment system **60** is

relatively compact in the vertical direction, and this makes it possible to install motors **56** of larger diameters.

(16) According to one configuration, the interface connection pins **74**, **76**, **82** are situated at a lower plane passing through the lower face **62** of the primary structure **58** or offset upward with respect to this lower plane. This arrangement makes it possible to further reduce the vertical bulk.

(17) According to one embodiment, the first interface connection pin **74** connected to the second end **70.2** of the first, L-shaped shackle **70** is positioned between the primary structure **58** and the second interface connection pin **76** connected to the intermediate zone **70.3** of the first shackle **70**.

(18) According to one arrangement, each first, L-shaped shackle **70** is connected to the right-hand interface **66** and each second shackle **78** is connected to the left-hand interface **68**. In a variant, each first, L-shaped shackle **70** could be connected to the left-hand interface **68** and each second shackle **78** could be connected to the right-hand interface **66**.

(19) Each first or second shackle **70**, **78** comprises, for each engine connection pin **72**, **80** and each interface connection pin **74**, **76**, **82**, a through-orifice, which houses the engine connection pin **72**, **80** or the interface connection pin **74**, **76**, **82**, dimensioned such that each engine or interface connection pin **72**, **74**, **76**, **80**, **82** is immobile in translation, in a transverse plane, with respect to the first or second shackle **70**, **78**.

(20) The first interface among the right-hand and left-hand interfaces **66**, **68** comprises, for each of the first and second interface connection pins **74**, **76**, at least one through-orifice, which houses the first or second interface connection pin **74**, **76**, dimensioned such that the first or second interface connection pin **74**, **76** is immobile in translation, in a transverse plane, with respect to the first interface. In addition, the second interface among the right-hand and left-hand interfaces **66**, **68** comprises at least one through-orifice, which houses the third interface connection pin **82**, dimensioned such that the third interface connection pin **82** is immobile in translation, in a transverse plane, with respect to the second interface. Thus, in normal operation, each interface connection pin **74**, **76**, **82** is configured to transmit the forces, at least in a transverse plane, between the right-hand and left-hand interfaces **66**, **68** and the first and second shackles **70**, **78**.

(21) The motor **56** comprises, for each engine connection pin **72**, **80**, at least one through-orifice, which houses the engine connection pin **72**, **80**, dimensioned such that each engine connection pin **72**, **80** is immobile in translation, in a transverse plane, with respect to the motor **56**. Thus, in normal operation, each engine connection pin **72**, **80** is configured to transmit the forces, at least in a transverse plane, between the motor **56** and the corresponding first or second shackle **70**, **78**.

(22) The right-hand interface **66** comprises at least one right-hand arm **66.1** positioned in a transverse plane and passed through by at least one of the first, second and third interface connection pins **74**, **76**, **82**. The left-hand interface **68** comprises at least one left-hand arm **68.1** positioned in a transverse plane and passed through by at least one of the first, second and third interface connection pins **74**, **76**, **82**.

(23) According to one configuration, the right-hand interface **66** comprises front and rear right-hand arms **66.1**, **66.2** positioned in transverse planes, parallel to each other and slightly spaced apart, and a right-hand base **66.3** connecting the front and rear right-hand arms **66.1**, **66.2**, the right-hand base **66.3** having a contact face **F66.3** pressed against the right-hand lateral face **64.1** of the primary structure **58** and being connected to the primary structure **58** by at least one connection element **66.4**, generally a plurality of connection elements **66.4** distributed in the vertical direction.

(24) The left-hand interface **68** comprises front and rear left-hand arms **68.1**, **68.2** positioned in transverse planes, parallel to each other and slightly spaced apart, and a left-hand base **68.3** connecting the front and rear left-hand arms **68.1**, **68.2**, the left-hand base **68.3** having a contact face **F68.3** pressed against the left-hand lateral face **64.2** of the primary structure **58** and being connected to the primary structure **58** by at least one connection element **68.4**, generally a plurality of connection elements **68.4** distributed in the vertical direction.

(25) In addition, the engine attachment system **60** comprises a single first shackle **70**, interposed

between the front and rear arms **66.1**, **66.2**, **68.1**, **68.2** of the right-hand or left-hand interface **66**, **68**, and a single second shackle **78** interposed between the front and rear arms **66.1**, **66.2**, **68.1**, **68.2** of the right-hand or left-hand interface **66**, **68**.

(26) Finally, the motor **56** comprises, for the first engine connection pin **72**, a first clevis **56.1** that has two arms between which the first shackle **70** is positioned and, for the second engine connection pin **80**, a second clevis **56.2** that has two arms between which the second shackle **78** is positioned.

(27) According to another configuration, each right-hand and left-hand interface **66**, **68** comprises a single right-hand or left-hand arm **66.1**, **68.1**. In addition, the engine attachment system **60** comprises two first shackles **70** positioned on either side of a first arm among the right-hand and left-hand arms and two second shackles **78** positioned on either side of a second arm, which is different from the first arm, among the right-hand and left-hand arms.

(28) Whatever the configuration, each right-hand or left-hand interface **66**, **68** has a contact face **66.3**, **68.3**, pressed against the right-hand or left-hand lateral face **64.1**, **64.2**, which does not extend against the lower face **62** and therefore beyond the right-hand or left-hand lateral face **64.1**, **64.2**.

(29) According to one embodiment, each right-hand or left-hand arm **66.1**, **66.2**, **68.1**, **68.2** has a substantially triangular contour having a first side **C1**, pressed against the primary structure **58**, which extends from the lower face **62** of the primary structure **58**, a second side **C2** positioned approximately in the continuation of the lower face **62** of the primary structure **58** and approximately in a horizontal plane, and a third side **C3**, the second and third sides **C2**, **C3** being connected by a rounded vertex at which is positioned the second interface connection pin **76** connected to the intermediate zone **70.3** of the first shackle **70** or the third interface connection pin **82** connected to the second end **78.2** of the second shackle **78**. The front and rear arms **66.1**, **66.2**, **68.1**, **68.2** of each right-hand or left-hand interface **66**, **68** have the same contour. Whatever the embodiment, each right-hand or left-hand arm **66.1**, **66.2**, **68.1**, **68.2** comprises a side oriented toward the motor **56**.

(30) According to one embodiment, the engine attachment system **60** comprises at least one reinforcing bracket **88** having a first wall **88.1** pressed against one of the right-hand or left-hand, front or rear arms **66.1**, **66.2**, **68.1**, **68.2** and a second wall **88.2** pressed against the right-hand or left-hand lateral face **64.1**, **64.2** and connected to the primary structure **58** by at least one connection element **88.3**, generally a plurality of connection elements **88.3** distributed in the vertical direction. Each first wall **88.1** comprises, for each interface connection pin **74**, **76**, **82** that passes through it, a through-orifice, which houses the interface connection pin **74**, **76**, **82**, dimensioned such that the interface connection pin **74**, **76**, **82** is immobile in translation, in a transverse plane, with respect to the first wall **88.1**. Thus, in normal operation, each of the first, second and third interface connection pins **74**, **76**, **82** is configured to transmit the forces, at least in a transverse plane, between the first wall **88.1** of the reinforcing bracket **88** and the first or second shackle **70**, **78** that are passed through by the first, second or third interface connection pin **74**, **76**, **82**.

(31) According to one configuration, the engine attachment system **60** comprises, for each front and rear arm **66.1**, **66.2**, **68.1**, **68.2** of each right-hand and left-hand interface **66**, **68**, a reinforcing bracket **88**, **88'**. Thus, for each right-hand and left-hand interface **66**, **68**, the engine attachment system **60** comprises a front reinforcing bracket **88**, positioned at the front of the right-hand or left-hand interface **66**, **68**, having a first wall **88.1** pressed against the front arm **66.1**, **68.1** of the right-hand or left-hand interface **66**, **68** and a second wall **88.2** pressed against the right-hand or left-hand lateral face **64.1**, **64.2** of the primary structure **58**, at the front of the right-hand or left-hand interface **66**, **68**, and a rear reinforcing bracket **88'**, positioned at the rear of the right-hand or left-hand interface **66**, **68**, having a first wall **88.1** pressed against the rear arm **66.2**, **68.2** of the right-hand or left-hand interface **66**, **68** and a second wall **88.2** pressed against the right-hand or left-hand lateral face **64.1**, **64.2** of the primary structure **58**, at the rear of the right-hand or left-hand interface

66, 68.

(32) According to one embodiment, each arm **66.1, 66.2, 68.1, 68.2** of each right-hand and left-hand interface **66, 68** comprises an extension **90** positioned at its second side **C2** oriented toward the motor **56** and offset toward the motor **56** with respect to the remainder of the second side **C2**, the extension **90** having a through-orifice **92** designed to house the right-hand or left-hand emergency connection pin **84, 86**.

(33) According to a first configuration, when the right-hand or left-hand interface **66, 68** comprises a single right-hand or left-hand arm **66.1, 68.1**, the motor **56** comprises, for each right-hand or left-hand emergency connection pin **84, 86**, a clevis that has two arms between which the extension **90** is at least partially positioned, each of the arms having a through-orifice designed to house the right-hand or left-hand emergency connection pin **84, 86**.

(34) According to a second configuration, when the right-hand or left-hand interface **66, 68** comprises front and rear arms **66.1, 66.2, 68.1, 68.2**, the motor **56** comprises, for each right-hand or left-hand emergency connection pin **84, 86**, a transverse flange **94** positioned at least partially between the front and rear arms **66.1, 66.2, 68.1, 68.2** of the corresponding right-hand or left-hand interface **66, 68**, the transverse flange **94** having a through-orifice designed to house the right-hand or left-hand emergency connection pin **84, 86**.

(35) Whatever the configuration, the right-hand or left-hand emergency connection pin **84, 86** provides a direct connection between each right-hand or left-hand interface **66, 68** and the motor **56** in the event of a failure. The engine attachment system **60** comprises, for each right-hand or left-hand emergency connection pin **84, 86**, a clearance between the right-hand or left-hand emergency connection pin **84, 86** and one element among the motor **56** and the right-hand or left-hand interface **66, 68**. This clearance is calibrated to allow displacement in a transverse plane between the motor **56** and the right-hand or left-hand interface **66, 68** during operation.

(36) According to a first arrangement, the right-hand or left-hand emergency connection pin **84, 86** is as one with the right-hand or left-hand interface **66, 68** and a clearance is provided between the right-hand or left-hand emergency connection pin **84, 86** and the motor **56**.

(37) According to a second arrangement, the right-hand or left-hand emergency connection pin **84, 86** is as one with the motor **56** and a clearance is provided between the right-hand or left-hand emergency connection pin **84, 86** and the right-hand or left-hand interface **66, 68**. Thus, according to the second configuration, the right-hand or left-hand emergency connection pin **84, 86** is as one with the transverse flange **94** of the motor **56** and a clearance is provided between the right-hand or left-hand emergency connection pin **84, 86** and each front and rear arm **66.1, 66.2, 68.1, 68.2** of the corresponding right-hand or left-hand interface **66, 68**.

(38) As illustrated in FIGS. 5 and 6, in normal operation, the first and second engine connection pins **72, 80** and the first, second and third interface connection pins **74, 76, 82** are configured to transmit the forces between the motor **56** and the right-hand and left-hand interfaces **66, 68** and optionally the reinforcing brackets **88, 88'**. In addition, the connection elements **66.4, 68.4** and **88.3** connecting the right-hand and left-hand interfaces **66, 68** and the reinforcing brackets **88, 88'** to the primary structure **58** are configured to transmit the forces between, on the one hand, the right-hand and left-hand interfaces **66, 68** and the reinforcing brackets **88, 88'** and, on the other hand, the primary structure **58**.

(39) As illustrated in FIG. 7, in the event of damage to a shackle **70, 78**, the right-hand or left-hand emergency connection pin **84, 86**, which is connected to the right-hand or left-hand interface **66, 68** to which the damaged shackle **70, 78** is connected, is configured to ensure, in addition to the undamaged shackle, the transmission of the forces between the motor **56** and the right-hand and left-hand interfaces **66, 68** and to compensate for this failure.

(40) As illustrated in FIG. 8A, in the event of failure of the right-hand or left-hand interface **66, 68**, the front and rear reinforcing brackets **88, 88'**, which are positioned on either side of the damaged right-hand or left-hand interface **66, 68**, are configured to compensate for this failure and to ensure

at least partial transmission of the forces between the motor 56 and the primary structure 58.

(41) As illustrated in FIG. 8B, in the event of failure of the front reinforcing bracket 88, the right-hand or left-hand interface 66, 68 and the rear reinforcing bracket 88', which are positioned against the same right-hand or left-hand lateral face 64.1, 64.2 of the primary structure 58 as the damaged front reinforcing bracket 88, are configured to compensate for this failure and to ensure at least partial transmission of the forces between the motor 56 and the primary structure 58.

(42) As illustrated in FIG. 8C, in the event of failure of the rear reinforcing bracket 88', the right-hand or left-hand interface 66, 68 and the front reinforcing bracket 88, which are positioned against the same right-hand or left-hand lateral face 64.1, 64.2 of the primary structure 58 as the damaged rear reinforcing bracket 88', are configured to compensate for this failure and to ensure at least partial transmission of the forces between the motor 56 and the primary structure 58.

(43) While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms "comprise" or "comprising" do not exclude other elements or steps, the terms "a" or "one" do not exclude a plural number, and the term "or" means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

1. An aircraft comprising: a motor, a primary structure of a pylon and an engine attachment system connecting the primary structure and the motor, said primary structure having a lower face and right-hand and left-hand lateral faces positioned on either side of a vertical median plane (PMV), the engine attachment system comprising: an L-shaped shackle, positioned in a transverse plane, which has first and second ends and an intermediate zone, a first engine connection pin connecting the first end of the first shackle and the motor, a second shackle, positioned in a transverse plane, which has first and second ends, a second engine connection pin connecting the first end of the second shackle and the motor, a right-hand interface connected to the right-hand lateral face of the primary structure, a left-hand interface connected to the left-hand lateral face of the primary structure, a first interface connection pin connecting the second end of the first shackle and a first interface among the right-hand and left-hand interfaces, a second interface connection pin connecting the intermediate zone of the first shackle and the first interface, a third interface connection pin connecting the second end of the second shackle and a second interface, which is different from the first interface, among the right-hand and left-hand interfaces, a right-hand emergency connection pin cooperating with the right-hand interface and the motor, a left-hand emergency connection pin cooperating with the left-hand interface and the motor, wherein each right-hand or left-hand interface comprises front and rear arms positioned in transverse planes, parallel to each other and slightly spaced apart, and a right-hand or left-hand base connecting the front and rear arms, said right-hand or left-hand base being connected to the primary structure by a connection element.

2. The aircraft as claimed in claim 1, wherein the first, second and third interface connection pins are situated at a lower plane passing through the lower face of the primary structure or offset upward with respect to this lower plane.

3. The aircraft as claimed in claim 1, wherein the first interface among the right-hand and left-hand interfaces comprises, for each of the first and second interface connection pins, a through-orifice, which houses the first or second interface connection pin, dimensioned such that the first or second

interface connection pin is immobile in translation, in a transverse plane, with respect to the first interface, and wherein the second interface among the right-hand and left-hand interfaces comprises a through-orifice, which houses the third interface connection pin, dimensioned such that the third interface connection pin is immobile in translation, in a transverse plane, with respect to the second interface.

4. The aircraft as claimed in claim 1, wherein the engine attachment system comprises a reinforcing bracket having a first wall pressed against one of the front and rear arms of the right-hand and left-hand interfaces and a second wall pressed against the right-hand or left-hand lateral face of the primary structure and connected to the primary structure by a connection element; the first wall having, for each of the first, second and third interface connection pins that passes through it, a through-orifice dimensioned such that the first, second or third interface connection pin is immobile in translation, in a transverse plane, with respect to the first wall.

5. The aircraft as claimed in claim 4, wherein, for each right-hand and left-hand interface, the engine attachment system comprises: a front reinforcing bracket, positioned at the front of the right-hand or left-hand interface, having a first wall pressed against the front arm of the right-hand or left-hand interface and a second wall pressed against the right-hand or left-hand lateral face of the primary structure, and a rear reinforcing bracket, positioned at the rear of the right-hand or left-hand interface, having a first wall pressed against the rear arm of the right-hand or left-hand interface and a second wall pressed against the right-hand or left-hand lateral face of the primary structure.

6. The aircraft as claimed in claim 1, wherein each front and rear arm of each right-hand and left-hand interface comprises a side oriented toward the motor and an extension positioned at said side and offset toward the motor with respect to a remainder of said side, said extension having a through-orifice that houses the right-hand or left-hand emergency connection pin.

7. The aircraft as claimed in claim 1, wherein the motor comprises, for each right-hand or left-hand emergency connection pin, a transverse flange positioned at least partially between the front and rear arms of the corresponding right-hand or left-hand interface, said transverse flange having a through-orifice that houses the right-hand or left-hand emergency connection pin.

8. The aircraft as claimed in claim 1, wherein the motor comprises, for the first engine connection pin, a first clevis that has two arms between which the first shackle is positioned and, for the second engine connection pin, a second clevis that has two arms between which the second shackle is positioned.

9. The aircraft as claimed in claim 1, wherein the engine attachment system comprises, for each right-hand or left-hand emergency connection pin, a clearance between the right-hand or left-hand emergency connection pin and one element among the motor and the right-hand or left-hand interface.

10. The aircraft as claimed in claim 1, wherein the motor comprises, for each of the first and second engine connection pins, a through-orifice, which houses the first or second engine connection pin, dimensioned such that each of the first and second engine connection pins is immobile in translation, in a transverse plane, with respect to the motor.
